

Fiona Prendergast

Software and Robotics Engineer

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EDUCATION

Worcester Polytechnic Institute — Worcester, MA

Aug 2022 — Present

B.S. Computer Science and Robotics Engineering — GPA 4.0/4.0

Anticipated Graduation May 2026

Academic Honors: Tau Beta Pi Engineering Honor Society, Dean's List (all semesters), Academic All-Conference (3 years)

Coursework: Software Engineering, AI & Machine Learning, Unified Robotics Series, Control Engineering, HCI & Human Robot Interaction, Database Systems I & II, Software Radio Design

SKILLS

C/C++, Python, Java, MATLAB, Git, Linux, Agile, AWS, ROS 2, Forward/Inverse Kinematics, Trajectory Planning, SLAM, Sensor Fusion, Kalman Filtering, ESP32, Arduino, Serial Communication, GPIO, Real-Time Systems, Circuit Design, Solidworks, CAD, Mechanical Systems, Advanced Spanish

WORK EXPERIENCE

Product Development Intern — Blueleaf Wealth Inc.

Jun 2023 — Present

- Built and deployed client financial app on live Expo Go deployment pipeline using React Native, implementing custom visual themes, graphics from live financial data API, industry standard authentication, and custom gesture interactions
- Designed a custom REST API for financial advisory software, using Postman to build a mock API for parallel development

Undergraduate Teaching Assistant — WPI Computer Science Department

Aug 2024 — Present

- Explained synchronization primitives, process scheduling, and memory management concepts during office hours
- Supported students debugging concurrent programs and understanding low-level system behavior

Integrations Software Engineering Intern — Vestmark, Inc.

Jun 2025 — Aug 2025

- Refactored legacy financial exchange code in Java, reducing cognitive complexity and achieving 100% unit test coverage
- Deployed an AWS cloud server migration for data adapter systems using EC2, VPN, and SFTP configurations

Embedded Systems Engineering Intern — DCS Corporation

Jun 2024 — Aug 2024

- Developed a serial data simulation tool for in-the-loop hardware testing to playback recorded navigation (GNSS) data
- Built a responsive Python GUI to interface simulation software with production hardware to simplify verification
- Wrote coverage-based unit tests from scratch and led code reviews to ensure simulation fidelity met verification requirements on a multi-million dollar defense project

PROJECTS

Robotic Parachute Packing System — Senior Capstone (MQP) Project

Present

- Architected a modular ROS 2 system with 5 custom packages featuring asynchronous action servers, service-based coordination, and configurable launch modes with real-time digital twin visualization
- Integrated dual stereo cameras, ESP32 microcontroller, and custom high-torque arm into a dual-agent control pipeline

Machine Learning Sports Injury Prediction Model — ML Final Project

Fall 2025

- Designed a multi-output regression model with end-to-end data preprocessing, feature selection, and cross-validation

Web-based Microfinance Platform — Ecuador based Research (IQP) Project

Spring 2025

- Built a web platform featuring an amortization calculator, document repository, and financial literacy resources
- Conducted user research with stakeholders of 20+ women's microfinance groups in Cuenca, Ecuador and Vice Mayor's office to validate platform requirements for accessibility

Autonomous Maze Navigation Robot — Unified Robotics Final Project

Spring 2025

- Programmed a TurtleBot in ROS to navigate unknown maze environments using SLAM, fusing lidar and odometry inputs
- Implemented a Kalman filter-based position estimation pipeline for accurate landmark-based navigation correction

LEADERSHIP & INVOLVEMENTS

- WPI Women's Soccer**, Team Captain

2022 - Present

- Student Athlete Advisory Committee**, Chair

2023 - Present

- Alpha Xi Delta Women's Fraternity**, Finance Director

2022 - Present

- Eagle Scout**, Founding member of Scouts BSA all-female Troop 65

Awarded Nov 2021